

- **Quandl:** Provides a wide range of financial and economic datasets (some free, some paid)

```
1. # pip install quandl
2. import quandl
3. quandl.ApiConfig.api_key = 'YOUR_API_KEY'
4. data = quandl.get("WIKI/AAPL")
```

- **IEX Cloud:** Offers high-quality stock data with a developer-friendly API

```
1. # pip install pyEX
2. import pyEX as px
3. c = px.Client(api_token='YOUR_API_KEY')
4. data = c.chartDF('AAPL')
5.
```

These services typically require creating an account and obtaining an API key, but many offer free tiers sufficient for learning and developing small-scale trading algorithms.

*?? **Pro Tip:** Professional traders maintain multiple data sources for the same asset classes. This redundancy isn't just for backup—it's for verification. When Two Sigma (a \$58 billion quantitative hedge fund) receives market data, their systems automatically compare it across different sources to identify and correct any discrepancies before feeding it to their trading algorithms. While this level of verification might be overkill for beginners, you can adopt a simplified version by occasionally comparing data between Yahoo Finance and another source like Alpha Vantage. If you spot significant differences, investigate before using that data for trading decisions.*

Cleaning and Preprocessing Financial Data

Raw financial data is rarely perfect. Missing values, outliers, and formatting issues can compromise your trading algorithms if not properly addressed.